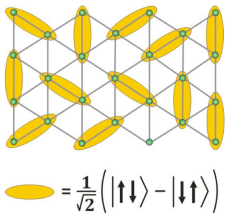
Sze Wai Pang

waipangsze@gmail.com
Hong Kong
github.com/waipangsze
waipangsze.github.io
orcid 0000-0002-4239-7566



Interests

- Physics, Mathematics, Numerical Weather Prediction (NWP), Data assimilation, Machine learning, Deep learning, Computational Fluid Dynamics (CFD)

Education

2013 – 2020 **PhD in Physics**
The Hong Kong University of Science and Technology (HKUST)
Theoretical and Computational Condensed Matter Physics

2010 – 2013 **Bachelor of Science in Physics**
The Hong Kong University of Science and Technology (HKUST)
Major: Physics and Mathematics

Employment

1/2024 – Now **Principal Scientist.** ClusterTech Limited
- Lead research teams, providing direction and guidance on project objectives and methodologies
- To design, implement, and oversee experiments aimed at product research and development
- Project management

7/2020 – 12/2023 **Senior Computational Scientist.** ClusterTech Limited
- Carry out the study of Numerical Weather Prediction (NWP) and Climate Change
- Analyse simulation results
- Collaborate with the academic sector and conduct research and development (R&D) for weather-related products/projects
- Support routine operations of forecasting systems as well as the underlying HPC/IT infrastructures

Research Publications

Journal Articles

1. Y. Zheng, C.-Y. Tam, X. Yu, J. Jiang, W.-P. Sze, and C.-C. Cheung, "A multi-model, multi-scheme evaluation of scale-aware convective parameterization for gray-zone tropical cyclone simulation over the south china sea," *Weather and Forecasting*, e250128, 2026. DOI: 10.1175/WAF-D-25-0128.1.
2. P.-H. Kam, C.-Y. Tam, W.-P. Sze, C.-C. Cheung, K.-K. Ng, and S.-H. Lee, "Dynamically adapting mesh refinement in an unstructured grid global model for numerical weather prediction," *Weather and Forecasting*, vol. 40, no. 8, pp. 1445–1462, 2025. DOI: 10.1175/WAF-D-24-0239.1.
3. W.-P. Sze, S.-C. Tang, C.-C. Cheung, and C.-Y. Tam, "Numerical weather prediction at 200 m local resolution based on an unstructured grid global model," *Earth and Space Science*, vol. 9, no. 10, e2022EA002342, 2022.

4 W. P. Sze, T. K. Ng, and K. T. Law, "Emerging ergodic behavior within many-body localized states," *arXiv preprint arXiv:2005.11812*, 2020.

Conference Proceedings

1 Chi-Chiu Cheung, Ka-Ki Ng, Wai-Pang Sze and Jimmy Tat Chi Wong, "MPAS-A with Hierarchical Timestepping and Customized Mesh Generation: 2023 Updates," Joint WRF/MPAS Users' Workshop 2023, 20 – 23 June, 2023, 2023. URL: https://www2.mmm.ucar.edu/wrf/users/workshops/WS2023/presentations/day3/1_cheung.pdf.

2 C.-C. Cheung, C.-Y. Tam, W.-N. Leung, K.-K. Ng, and W.-P. Sze, "Applications of flexible spatial and temporal discretization techniques to a numerical weather prediction model," in *Proceedings of the Platform for Advanced Scientific Computing Conference*, ser. PASC '22, Basel, Switzerland: Association for Computing Machinery, 2022, ISBN: 9781450394109. DOI: 10.1145/3539781.3539790.

Skills

Languages	English, Mandarin Chinese, Cantonese
Coding	Python, FORTRAN, Shell Script, C++, Matlab, NCL, LaTeX
Software	WRF, MPAS, JEDI-MPAS(DA), ecFlow, PALM, OpenFOAM, Conda/Micromamba, Spack, Singularity, SLURM, Pytorch
Misc.	Academic research, Teaching, Training, Consultation

Miscellaneous Experience

Works

Technical

- Performed sensitivity analyses on atmospheric models (WRF/MPAS/CPAS), optimizing physics schemes and applying results to real-world business projects
- Carried out model tuning, including mesh design and adjustments to PBL (YSU and scale-aware Shin-Hong), GWDO (YSU and scale-aware GSL), cloud fraction, and surface drag parameterization through namelist options (topo_wind)
- Integrated WRF-UCM (Urban Canopy Model) into the MPAS/CPAS framework and applied it to various projects
- Integrated MPAS/CPAS with OpenFOAM to provide meteorological inputs for CFD simulations
- Managed and maintained operational meteorological simulations using the CPAS model via ecFlow
- Developed and optimized JEDI-MPAS for enhanced performance
- Applied machine learning for data analysis and post-processing, including bias correction

Business

- Participated in multiple Exhibitions including 2022 LOGTECH, 2023 International Maritime Conference and Exhibition (Marintec), 2023 InnoEX, 2024 Singapore Asia Climate Forum, and 2024 Nanjing SmartHK Jiangsu-Hong Kong High-Quality Development Cooperation Conference
- Client visits
- Prepared Tender Documents, including technical proposals and cost estimations, and participated in tender interviews, such as presenting solutions at the Meteorological Service Singapore (MSS) office

Miscellaneous Experience (continued)

Projects

2023 - now

CUHK: Climate Dynamic Downscaling Simulation
Project Title: CPAS Model Climate Dynamic Downscaling Simulation
• To conduct high-resolution, multi-year meteorological and environmental simulations using advanced modeling techniques to analyze historical trends and future climate impacts in the East Asia, South China, and GBA regions.

2022 - 2024

Innovation and Technology Fund - Enterprise Support Scheme
Project Title: Extreme Weather Prediction for Mega-Cities based on the Global Model CPAS: Enabling Technologies and Delivery
• To develop an advanced system for predicting and mitigating the impacts of extreme weather events in the world's largest metropolitan areas.

2020 - 2022

Innovation and Technology Fund - Enterprise Support Scheme
Project Title: High Resolution Numerical Weather Simulation: A commercial service platform for commercial/public sectors, with a sample application on pollution control
• To develop a cutting-edge commercial service platform that provides high-resolution numerical weather simulation capabilities

2013 - 2020

PhD project
Supervisor: Prof. Tai Kai Ng (HKUST)
Project Title: Strongly correlation system and related numerical study
• A Many-Body Ergodic Phase between Two Classes of Many-Body Localized States and The Variational Study on Triangular XXZ Model

2016 - 2017

Projective Symmetry Group (PSG) Project
Collaborator: Prof. Zheng-Xin Liu (Department of Physics, Renmin University)
Project Title: Investigation of the classification problem of spin liquid state on XXZ triangular lattice
• In the previous work, I had finished the Variational Monte Carlo simulation about the spin-liquid type state. And then, Prof. Liu invited me to be a visitor and investigate the classification problem of spin liquid state on XXZ triangular lattice. The main idea comes from XG Wen's paper called Projective Symmetry Group(PSG). It is mainly theoretical work. Finally, we had done the classification analysis. (6/2016-8/2016 and 3/2017-4/2017)

3/2016 – 5/2016

Variational Monte Carlo (VMC) Project
Supervisor: Prof. Yi Zhou (Department of Physics, Zhejiang University)
Project Title: Optimization the trial wavefunction of spin liquid of 1D XXZ+DM spin model
• The project of spin liquid state by Variational Monte Carlo simulation. It is about the 1D XXZ+DM spin model and mainly focused on anisotropic spin spin interaction. Second, use *Tianhe-1* for simulation.

Miscellaneous Experience (continued)

2012 - 2013 **Final Year Project**
Supervisor: Prof. Tai Kai Ng (HKUST)
Project Title: Superfluid-insulator transition
• The phase diagrams and phase transitions of Bosons with short-ranged repulsive interactions moving in periodic or random external potentials at zero temperature are investigated with emphasis on the superfluid-insulator transition induced by varying a parameter such as the density

2011 – 2012 **Undergraduate Research Opportunities Program (UROP)**
Supervisor: Prof. Kwok Yip SZETO (HKUST)
Project Title: Prisoner Dilemma Game on Complex Networks
• The prisoner dilemma game is generalized in two aspects in the multi-agent system framework. They are agents have memory and the interaction between agents are defined on a social network. My work focuses on Wheatstone Network and no memory

References

Available on Request